

Roll: _____

Name: _____

[Q1] (MCQ+1) Assume a uniform-cost search (UCS) on a tree with branching factor of b , if the cost function $g(n)$ is linear, $g(n) = k \times d(n)$ where k is a constant and $d(n)$ is the depth of node n , how many nodes are expanded by UCS before it finds the optimal solution at depth δ starting from the root-node?

(Note: depth of root node is 0)

[A] at least: $(k\delta - 1)/(b - 1)$

[B] at least: $(b^\delta - 1)/(b - 1)$

[C] at most: $(b\delta - 1)/(b - 1)$

[D] at most: $b(\delta^b - 1)/(b - 1)$

[Q2] (MCQ+1) Assume an A* search algorithm is used on a graph where the heuristic function $h(n)$ is admissible. If the cost of the optimal path from the start node S to the goal node G is C^* , what is the maximum possible cost F of the first solution found by A* search in the worst case?

[A] $F > 2C^*$

[B] $F > C^*$

[C] $F = C^*$

[D] $F = C^* + h(G)$

[Q3] (MSQ+2) The heuristic path algorithm is a best-first search in which $f(n) = (2 - w)g(n) + wh(n)$. Which of the following statements are true in this context?

[A] for $w = 1$, $f(n)$ represents the A* algorithm

[B] for $w = 2$, $f(n)$ is complete

[C] for $w > 2$, $f(n)$ is optimal

[D] for $w = 0$, $f(n)$ represents UCS

[Q4] (MSQ+2) Select the *correct* statement from the following:

[A] DFS is complete but not optimal

[B] If h is an admissible heuristic (non-negative), then $h + 1$ can *never* be an admissible heuristic

[C] if x and y are admissible heuristics then $\alpha x + (1 - \alpha)y$ is guaranteed to be admissible for $0 < \alpha < 1$

[D] if x and y are two admissible heuristics, then $\max(x, y)$ dominates both x and y

[Q5] (MSQ+2) Select the evaluation function that will result in identical behavior to greedy best-first search.

[A] $f(n) = h(n)/2$

[B] $f(n) = h(n)$

[C] $f(n) = \sqrt{h(n)}$

[D] $f(n) = g(n) * h(n)$

[Q6] (MSQ+2) Select the *correct* statements with regards to the minimax search algorithm?

[A] It is optimal for all kinds of adversaries

[B] It is complete if the search tree is finite

[C] with pruning, it can potentially lead to suboptimal solutions

[D] the order in which nodes are visited affects the amount of pruning

[Q7] (MSQ+2) In minimax search with pruning, consider a player who had a chance of moving to a node n somewhere in the search tree, the player will choose not to move to this node when

[A] It has a better choice at the same depth

[B] It had a better choice at some lower depth

[C] It has a better choice at some higher depth

[D] If the other player has a better choice at the same depth

In the Wumpus World, the environment can be represented using propositional logic. Let's define the following:

- $P(i, j)$: There is a pit in cell (i, j) .
- $W(i, j)$: The Wumpus is in cell (i, j) .
- $B(i, j)$: There is a breeze in cell (i, j) .
- $S(i, j)$: There is a stench in cell (i, j) .
- $L(i, j)$: Agent is in the cell (i, j) .
- $A(i, j)$: The agent is in a cell adjacent to (i, j) .

Answer the following questions based on the above premise.

[Q8] (MCQ+2) We would like to define a logical rule which says: - *If and only if there is a pit in cell (i, j) , then cells adjacent to (i, j) will have a breeze.* Select the corresponding rule expressed in terms of propositional logic statement:

- [A] $P(i, j) \leftrightarrow (B(i-1, j) \wedge B(i+1, j) \wedge B(i, j-1) \wedge B(i, j+1))$
- [B] $P(i, j) \leftrightarrow (B(i-1, j) \vee B(i+1, j) \vee B(i, j-1) \vee B(i, j+1))$
- [C] $(B(i-1, j) \vee B(i+1, j) \vee B(i, j-1) \vee B(i, j+1)) \rightarrow P(i, j)$
- [D] $(B(i-1, j) \wedge B(i+1, j) \wedge B(i, j-1) \wedge B(i, j+1)) \rightarrow P(i, j)$

[Q9] (MCQ+2) Based on the rule defined in the previous question, consider a cell (i, j) where a breeze is perceived by the agent. Which of the following statements is logically equivalent to this perception?

- [A] $\exists (k, l)$ such that $(P(k, l) \vee A(i, j))$
- [B] $\exists (k, l)$ such that $(P(k, l) \wedge A(k, l))$
- [C] $\exists (k, l)$ such that $(P(k, l) \wedge L(k, l))$
- [D] $\exists (k, l)$ such that $(P(k, l) \vee L(i, j))$

[Q10] (MCQ+2) Assume that the wampus, instead of stenching itself, is attracted to naturally occurring stench in the world. Assume that there is only one cell in the world containing a stench. Consider the statement:- *if a cell has a stench, then the Wumpus must be nearby (in the cell or in one of the adjacent cells).* Which of the following statements represent this proposition correctly?

- [A] $S(i, j) \leftrightarrow (W(i-1, j) \vee W(i+1, j) \vee W(i, j-1) \vee W(i, j+1) \vee W(i, j))$
- [B] $S(i, j) \leftrightarrow (W(i-1, j) \wedge W(i+1, j) \wedge W(i, j-1) \wedge W(i, j+1) \wedge W(i, j))$
- [C] $S(i, j) \rightarrow (W(i-1, j) \vee W(i+1, j) \vee W(i, j-1) \vee W(i, j+1) \vee W(i, j))$
- [D] $S(i, j) \rightarrow (W(i-1, j) \wedge W(i+1, j) \wedge W(i, j-1) \wedge W(i, j+1) \wedge W(i, j))$

[Q11] (MCQ+1) Based on the rules defined in the previous questions, given that there is no stench perceived in cell (i, j) , which of the following statements is entailed?

- [A] There is a pit in at least one of the adjacent cells.
- [B] There is a stench in at least one of the adjacent cells.
- [C] The wumpus is not in any of the adjacent cells.
- [D] None of the above.

[Q12] (MCQ+1) Which of the following statements correctly reflects the logical entailment given that there is a pit in cell (i, j) ?

- [A] "There is no breeze in any of the adjacent cells" is true.
- [B] "There is a breeze in at least one adjacent cell" is true.
- [C] "There is a stench in cell (i, j) " is true.
- [D] "Agent is adjacent to cell (i, j) " is true.

NOTE:

1. Write your **roll** number and **name** on the question paper.
2. Questions are either **MCQ**(single correct answer) or **MSQ**(multiple correct answers) as indicated alongside.
3. Partial marks will be awarded to partially correct answers for MSQs. However, no marks will be awarded if any incorrect option is chosen for that MSQ.
4. Mark the options of your choice **clearly** (avoid overwriting and avoid overlapping with other options).